Client Recommendations Report

Metric	Logistic Regression	Random Forest
Accuracy	84.58%	89.66%
Precision	41.81%	55.23%
Recall	81.10%	61.34%
F1-score	55.18%	58.13%
ROC-AUC	90.97%	92.66%

Executive Recommendation

For the current experimental dataset, Random Forest is the preferred model for further development. It has outperformed Logistic Regression in accuracy (89.66%), precision (55.23%), F1-score (58.13%), and ROC-AUC (92.66%). However, Logistic Regression achieved substantially higher recall (81.10% versus 61.34%), identifying more actual subscribers. The operational choice therefore depends on whether missed opportunities are more costly than unnecessary customer interventions. The McNemar test confirmed that the models produced significantly different classifications on the same test observations (test statistic 217.3720; p approximately 3.39 x 10^-49).

1. **Recommended model**

Random Forest is the stronger candidate for this experiment because it achieved more balanced overall performance. Compared with Logistic Regression, its accuracy was 5.08 percentage points higher at 89.66%, and its precision was 13.42 percentage points higher. It also recorded slightly better F1-score and ROC-AUC results. The confusion matrices further support this choice: Random Forest generated 526 false positives versus 1,194 for Logistic Regression. This advantage matters when positive predictions trigger sales or service activity, as fewer false positives reduce unnecessary operational effort.

This recommendation is conditional. Logistic Regression identified 858 of the 1,058 actual subscribers in the test set, compared with 649 for Random Forest. If the client's priority is to identify as many potential positive customers as possible, Logistic Regression's higher recall may support using a different threshold or operating strategy. Random Forest should therefore be viewed as the preferred development candidate, not as universally superior.

2. **Improving the selected model**

First, remove or test the call-duration variable separately. Duration was the most influential Random Forest feature, with an importance of approximately 0.321, but it is unavailable when predictions must be made before customer contact. Both models should therefore be retrained without duration, and the resulting changes in precision, recall, F1-score and ROC-AUC should be reported. This would provide a more realistic estimate of deployable performance.

Second, tune the Random Forest hyperparameters systematically instead of relying on the initial configuration. Cross-validation should test the number of trees, tree depth, minimum samples per split and leaf, and the number of features considered at each split. The probability threshold should also be tuned because the default of 0.50 may not reflect the client's commercial costs of false positives and false negatives. A more suitable threshold can be selected using validation data and an agreed business objective.

Third, include ongoing model monitoring in any production deployment. Regularly assess shifts in class balance, predictor distributions and error rates. Treat feature importance as a diagnostic rather than proof of causality and use out-of-time validation when sufficient historical client data are available.

3. Adapting the model to the requested SS1 client data

The public UCI Bank Marketing dataset serves only as an experimental proxy and predicts term-deposit subscriptions. For the client application, the modelling pipeline must instead use the outcome specified in the SS1 data request rather than assume that bank subscriptions are equivalent to the client's activation outcome. The first production step is therefore to define the target clearly, including what counts as a successful activation, the observation date and the prediction horizon.

The SS1 fields should then be mapped to the modelling schema. Variables with comparable business meaning can be retained, while UCI-specific variables without client equivalents should be removed or replaced with suitable client predictors. Categorical levels, missing values, numeric ranges and class balance must be reassessed because preprocessing developed for the public dataset may not suit the client data. Feature engineering should also be reviewed to ensure that derived variables are relevant to STADIOEquities and available when predictions are made.

The chosen model must then be retrained and validated using historical client data. The train/test split should reflect the intended prediction setting, with an out-of-time test preferred when dates are available to better assess future performance. Data governance controls should ensure that only authorised, relevant fields are used and that prediction outputs connect to an appropriate client workflow.

4. Alignment with the literature

The findings broadly support the Part A literature: machine-learning methods can identify useful predictive patterns in bank marketing data. Moro, Cortez and Rita (2014) used data-driven modelling to predict bank telemarketing success, while Hou et al. (2022) examined machine-learning models for bank deposit subscriptions. Similarly, both models in this experiment showed strong discrimination, with ROC-AUC values above 0.90.

The comparison also reinforces the literature's broader point that model selection and predictor design influence results. Random Forest and Logistic Regression produced materially different error profiles, and the McNemar test confirmed a statistically significant difference in their paired classifications. Duration's strong influence also shows why predictive performance must be interpreted considering when a variable becomes available.

The findings therefore indicate that the modelling approach is promising, but not that the public-data model can be transferred unchanged to STADIOEquities.

Conclusion

Random Forest should remain the primary candidate because it achieved higher accuracy, precision, F1-score and ROC-AUC on the public dataset. Logistic Regression should be retained as a benchmark because of its substantially higher recall. Before deployment, the preferred model should be tested without predictors unavailable at decision time, tuned through cross-validation and business-sensitive thresholds, and retrained on the requested SS1 client data using the client's actual definition of activation. This approach retains the value of the experiment while accounting for differences between the public benchmark and the intended client application.

References

1. Hou, S., Cai, Z., Wu, J., Du, H. and Xie, P. (2022) 'Applying machine learning to the development of prediction models for bank deposit subscription', *International Journal of Business Analytics*, 9(1), pp. 1–14. doi: 10.4018/IJBAN.288514.
2. Moro, S., Cortez, P. and Rita, P. (2014) 'A data-driven approach to predict the success of bank telemarketing', *Decision Support Systems*, 62, pp. 22–31. doi: 10.1016/j.dss.2014.03.001.
3. Moro, S., Rita, P. and Cortez, P. (2014) *Bank Marketing* [Dataset]. UCI Machine Learning Repository. doi: 10.24432/C5K306.